
Metric Panoramic Depth Estimation for Open-World

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1 What research problem are you aiming to solve?

This project aims to solve the challenge of **panoramic metric depth estimation** that generalizes across diverse indoor and outdoor environments. While perspective depth estimation has seen significant breakthroughs, panoramic depth estimation still struggles to achieve robust, metrically consistent predictions in real-world scenes, particularly outdoors Piccinelli et al. (2025); Guo et al. (2025); Cao et al. (2025). We aim to build a foundation model, named **DAP (Depth Any Panoramas)**, that provides absolute distance information for 360° environments regardless of the scene type.

2 Why is this problem important?

Panoramic depth estimation is critical for various "Spatial Intelligence" applications Lin et al. (2025):

- **Robotics:** It enables omnidirectional obstacle avoidance and navigation for autonomous agents like drones Zhang et al. (2025).
- **Metric Accuracy:** Unlike relative depth, metric depth provides actual physical distances, which is the cornerstone for 3D reconstruction and precise localization Li et al. (2025).
- **Data Scarcity:** Existing annotated panoramic datasets are limited in scale and diversity due to high collection costs Piccinelli et al. (2025); Cao et al. (2025); Li et al. (2025). Solving this problem allows for the utilization of massive unlabeled web-scale data.

3 Existing works and their limitations

Existing approaches to panoramic depth can be categorized into:

- **In-domain/Zero-shot Methods:** Models like Panda, DA^2 , and Depth Anywhere often rely on limited datasets (usually around 100k–800k samples) Cao et al. (2025).
- **Unified Frameworks:** Methods such as DAC and Unik3D attempt to unify camera geometries but often exhibit over-smoothed boundaries and fail to generalize to distant outdoor regions Guo et al. (2025); Piccinelli et al. (2025).
- **Limitations:** Most existing models struggle with Equirectangular Projection (ERP) distortions, particularly at the poles, and lack the scalability to handle diverse real-world textures and lighting conditions Li et al. (2025); Jiang et al. (2021); Wang and Liu (2024).

4 Proposed solution

We propose a scalable data-in-the-loop paradigm consisting of three main pillars:

1. **Large-scale Data Engine:** We construct a dataset of over **2 million samples**, combining 1.7M unlabeled real panoramas with 90k high-quality synthetic outdoor samples from the AirSim360 simulator Ge et al. (2025).

2. Three-Stage Pipeline:

- **Stage 1:** Train a Scene-Invariant Labeler on high-quality synthetic data.
 - **Stage 2:** Use a PatchGAN-based discriminator to filter 600k high-confidence real samples to train a Realism-Invariant Labeler, bridging the synthetic-to-real gap.
 - **Stage 3:** Final DAP training on the full 2M dataset.
3. **Geometry-Consistent Network:** Using DINOv3-Large as a backbone, we introduce a **plug-and-play range mask head** to adapt to varying distances (10m to 100m) and a multi-loss optimization strategy.

5 Novelty and significance

The novelty of our solution lies in:

- **Unprecedented Scale:** This is the first panoramic foundation model trained on a 2M+ sample scale.
- **Advanced Loss Functions:** We employ **Sharpness-centric** ($\mathcal{L}_{DF}, \mathcal{L}_{grad}$) and **Geometry-centric** ($\mathcal{L}_{normal}, \mathcal{L}_{pts}$) losses combined with a distortion-aware map to ensure structural fidelity across the sphere.
- **Superior Generalization:** DAP achieves state-of-the-art zero-shot performance on Stanford2D3D, Matterport3D, and Deep360, significantly reducing AbsRel error compared to previous methods.

6 Tentative timeline

The project will be completed within a two-month (8-week) period:

Week	Key Tasks
Week 1-2	Data collection, synthetic data generation with AirSim360, and environment setup.
Week 3-4	Training Stage 1: Scene-Invariant Labeler development and evaluation.
Week 5	Discriminator-based data filtering and pseudo-label generation for 1.7M images.
Week 6	Training Stage 2: Refining labels with the Realism-Invariant Labeler.
Week 7	Training Stage 3: Full DAP model training with range mask heads and geometric losses.
Week 8	Evaluation on benchmarks (DAP-Test) and final project reporting.

References

- Zidong Cao, Jinjing Zhu, Weiming Zhang, Hao Ai, Haotian Bai, Hengshuang Zhao, and Lin Wang. Panda: Towards panoramic depth anything with unlabeled panoramas and mobius spatial augmentation. In *CVPR*, 2025.
- Xian Ge, Yuling Pan, Yuhang Zhang, Xiang Li, Weijun Zhang, Dizhe Zhang, Zhaoliang Wan, Xin Lin, Xiangkai Zhang, Juntao Liang, et al. Airlsim360: A panoramic simulation platform within drone view. *arXiv preprint arXiv:2512.02009*, 2025.
- Yuliang Guo, Sparsh Garg, S Mahdi H Miangoleh, Xinyu Huang, and Liu Ren. Depth any camera: Zero-shot metric depth estimation from any camera. In *CVPR*, 2025.
- Hualie Jiang, Zhe Sheng, Siyu Zhu, Zilong Dong, and Rui Huang. Unifuse: Unidirectional fusion for 360 panorama depth estimation. *RAL*, 2021.
- Haodong Li, Wangguangdong Zheng, Jing He, Yuhao Liu, Xin Lin, Xin Yang, Ying-Cong Chen, and Chunchao Guo. Da²: Depth anything in any direction. *arXiv preprint arXiv:2509.26618*, 2025.
- Xin Lin, Xian Ge, Dizhe Zhang, Zhaoliang Wan, Xianshun Wang, Xiangtai Li, Wenjie Jiang, Bo Du, Dacheng Tao, Ming-Hsuan Yang, et al. One flight over the gap: A survey from perspective to panoramic vision. *arXiv preprint arXiv:2509.04444*, 2025.

- Luigi Piccinelli, Christos Sakaridis, Mattia Segu, Yung-Hsu Yang, Siyuan Li, Wim Abbeloos, and Luc Van Gool. Unik3d: Universal camera monocular 3d estimation. In *CVPR*, 2025.
- Ning-Hsu Albert Wang and Yu-Lun Liu. Depth anywhere: Enhancing 360 monocular depth estimation via perspective distillation and unlabeled data augmentation. *NeurIPS*, 2024.
- Qiang Zhang, Zhang Zhang, Wei Cui, Jingkai Sun, Jiahang Cao, Yijie Guo, Gang Han, Wen Zhao, Jiaxu Wang, Chenghao Sun, et al. Humanoidpano: Hybrid spherical panoramic-lidar cross-modal perception for humanoid robots. *arXiv preprint arXiv:2503.09010*, 2025.